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ORIGINAL SCIENTIFIC PAPER
Acute Coronary Syndromes

Impact of high-grade atrioventricular block on outcomes in patients with acute myocardial infarction

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Aims

Contemporary data on the impact of high-grade atrioventricular blocks (HAVB) in acute myocardial infarction (AMI) are scarce. Therefore, we investigated this in the Swiss national AMI registry (AMIS Plus) cohort.

Methods and results

We included all AMI patients enrolled from January 2005 to September 2024 with available ECG information. Primary outcome was in-hospital mortality; secondary outcomes included in-hospital major adverse cardiac or cerebrovascular events (MACCE) defined as cardiogenic shock, stroke, re-infarction or death. Among 50 279 AMI patients, 747 (1.5%) patients presented with HAVB and 407 (0.8%) developed HAVB during hospitalization. Patients presenting with HAVB had increased in-hospital mortality (15% vs. 4.9%; $P < 0.001$) and MACCE (17% vs. 6.0%; $P < 0.001$) compared with those without HAVB. High-grade atrioventricular blocks on admission was an independent predictor for in-hospital mortality (OR 1.89; 95% CI, 1.42–2.49; $P < 0.001$), with a stronger impact on mortality in the setting of anterior compared to non-anterior AMI (OR 3.69; 95% CI, 1.99–6.81; $p_{interaction} = 0.02$). Independent predictors for HAVB on admission included STEMI and female sex. Rates of permanent pacemaker implantation during hospitalization in patients with HAVB on admission were higher in NSTEMI than STEMI patients (20% vs. 4.4%; $P < 0.001$). High-grade atrioventricular blocks occurring during hospital stay was associated with higher mortality (20% vs. 15%; $P = 0.031$) and higher permanent pacemaker rate (25% vs. 8%; $P < 0.001$) than HAVB on admission.

Conclusion

High-grade atrioventricular blocks in AMI was associated with high in-hospital mortality, especially in anterior AMI or if occurring during hospitalization. Permanent pacemaker implantation was more frequent in the setting of NSTEMI and among patients developing HAVB during hospital stay.

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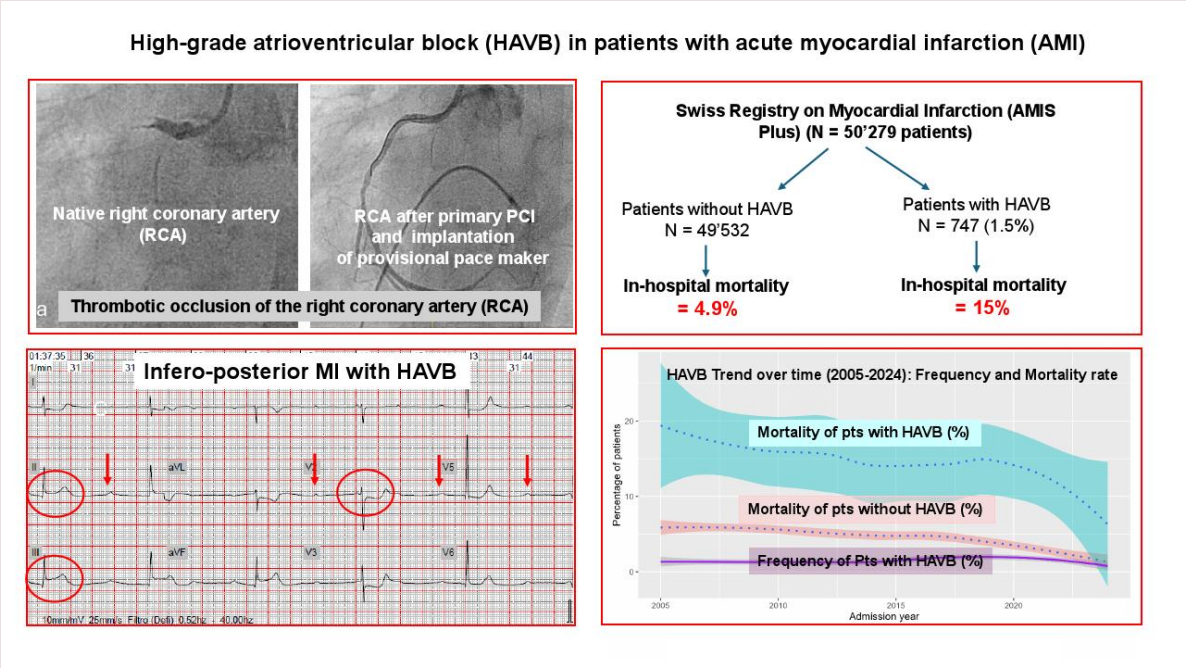
Graphical abstract



Keywords High-grade atrioventricular block • Acute myocardial infarction • Complications • Acute coronary syndromes

Background

High-grade atrioventricular block (HAVB), defined as second- or third-degree AV block, has decreased from ~20% in the thrombolytic era[1] to contemporary 2–3% in patients presenting with acute myocardial infarction (AMI).[2–4] High-grade atrioventricular block has been more frequently described in STEMI than in NSTEMI[2,4–6] and primarily in patients with inferior infarction location.[2,6,7]

High-grade atrioventricular block in the setting of AMI has been associated with an in-hospital mortality in the range of 20%,[2,6] but contemporary data are scarce. The purpose of this study is to address the impact of HAVB on in-hospital outcomes in patients presenting with AMI in the large Swiss National Registry on AMI (AMIS Plus).

Methods

Study design and objective

Acute Myocardial Infarction SwitzerlandPlus is an ongoing, prospective national registry in Switzerland that has been collecting data on AMI patients since 1997 (NCT01305785). All 84 participating Swiss hospitals have collected data for the registry using a standardized questionnaire collecting data on clinical presentation, medical treatment, in-hospital outcomes and complications. The AMIS Plus Data Centre, located at the Epidemiology, Biostatistics and Prevention institute of the University of Zürich, oversees data collection as well as the controlling for completeness and consistency.[8] The Swiss Federal Ethics Committee for Clinical Studies, the Swiss Board for Data Security and the corresponding cantonal ethic committees gave their approval for the registry. The present study is in accordance with the ethical guidelines of the Declaration of Helsinki.

Study population

Patients with either ST- or non-ST-elevation MI (STEMI, NSTEMI) enrolled in the registry between January 2005 and September 2024 with available electrocardiogram (ECG) data were included for the present study. Acute myocardial infarction was defined following the valid Universal Definition of Myocardial Infarction at the time of patient's admission to the registry.[9–12] The study inclusion is illustrated in *Figure 1*.

According to the AMIS Plus protocol, HAVB refers to second-degree type II (Mobitz II) block, high-risk 2:1 AV block, and third-degree AV block, conditions that may require provisional pacemaker implantation. This definition does not include suprahisian benign AV blocks. Patients were analysed based on HAVB presence at admission while its occurrence during hospitalization was also tracked as complication of AMI. Unfortunately, our dataset does not include precise timing data on the onset of HAVB, which means we cannot differentiate between conduction blocks related to the treatment intervention itself and those resulting from infarct progression.

Data collection

Data on baseline characteristics, clinical presentation, procedural characteristics, in-hospital treatment, and in-hospital outcomes were collected for all included patients. The classification system from 1967 by Killip and Kimball[13] was applied to estimate the clinical status at admission. Cardiovascular risk factors (CVRFs) such as hypertension, dyslipidaemia, and diabetes mellitus were considered to be present when the patient had previously been treated for this condition and/or was diagnosed by a medical doctor. Patients were classified as current smokers if they had at least smoked 100 cigarettes in their life and were currently smoking at the time of AMI. A body mass index (BMI) over 30 kg/m^2 was defined as obesity. The Charlson Comorbidity Index (CCI)[14,15] was calculated to assess the burden of comorbidities. Left ventricular ejection fraction (LVEF)

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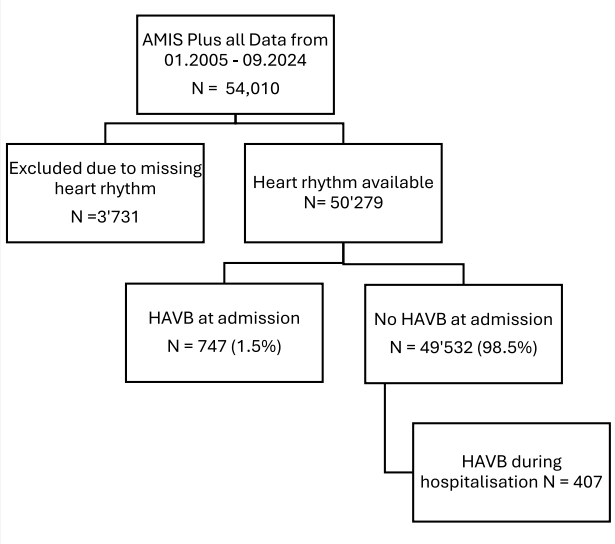

Figure 1 Flowchart for study inclusion of patients.

was defined as impaired when the angiographic measurement was below 35% or the echocardiographic assessment was below 30%.

Complications during hospitalization were acute kidney failure diagnosed if dialysis was necessary; a new heart failure was defined as a Killip class above II; cardiogenic shock was defined as hypotension (systolic blood pressure <90 mmHg) and clinical signs of hypoperfusion including cyanosis, cold extremities, and/or oliguria[16]; cerebrovascular events included any neurological event confirmed by a neurologist and/or an adequate imaging modality[17]; and reinfarction was defined as the presence of recurrent symptoms suggestive for ischaemia with concomitant characteristic ECG changes and/or a newly raise of cardiac biomarkers of at least two-fold above the upper limit of normal levels.[18]

Outcome measures

The primary outcome measure was in-hospital mortality. Secondary outcome measure was in-hospital MACCE, defined as the composite of cardiogenic shock, cerebrovascular events, re-infarction, or death.

Statistical analysis

Continuous variables were displayed as medians with interquartile range (IQR) while categorical variables as percentages. The Wilcoxon rank-sum test was used to com-pare the continuous variables. Categorical variables were compared using Pearson's Chi-squared test or Fisher's exact test. Initially performed univariate analysis compared patients regarding the presence of HAVB at admission. Patients with high-grade atrioventricular block were further stratified and compared regarding their diagnosis of AMI (STEMI vs. NSTEMI) or the onset of HAVB (at admission vs. during hospitalization). A multivariable logistic regression model was applied to identify independent predictors of HAVB in acute MI patients at admission considering the variables age, gender, diagnosis (STEMI/NSTEMI), Killip > 2, resuscitation prior admission, ischaemic region (anterior, inferior, posterior), diabetes mellitus, CCI > 1 and current smoker status. The assessment of the adjusted association between HAVB and in-hospital mortality was performed likewise, using a multivariable logistic regression analysis. The latter was adjusted for the same covariates (except current smoker) as in the analysis for predictors of HAVB. Additionally, it was adjusted for treatment with primary percutaneous coronary intervention (pPCI). To test if the impact of HAVB on in-hospital mortality is stronger in patients with an anterior MI, the model was repeated including an interaction term between HAVB and anterior region. Covariates were defined according to significant differences from the univariate analysis and previously described predictors for HAVB[2,4,7,19,20] or in-hospital mortality[6,21–23] in AMI patients. Results of these analyses are presented as odds ratios, 95% confidential intervals (CIs) and *P*-values.

Linear temporal trends were assessed by smoothed line plots and the Mantel–Haenszel test of trend to display eventual changes in the prevalence of HAVB in AMI patients and mortality rates over the past years.

A *P*-value of 0.05 or below was considered statistically significant. All described analysis was performed using R (Version 4.3.1.)

Results

Patient demographics and clinical characteristics

From 2005 to 2024, 54 010 patients were enrolled in the registry and 50 279 patients (93%) with available ECG information were included to this study. The median age of the patients was 66.5 years (IQR: 56.6–76.6 years), 74% were male and 57% presented with STEMI. Overall, HAVB at admission was observed in 747 patients (1.5%).

Baseline characteristics of AMI patients with or without HAVB at admission are displayed in *Table 1*. Patients with high-grade atrioventricular blocks were ~6 years older than patients without HAVB, had with a higher proportion of females (2.1% vs. 1.3%; $P < 0.001$) and were more frequently diabetic (26% vs. 21%; $P = 0.001$), multimorbid (defined as CCI score > 1: 27% vs. 23%; $P = 0.007$) and more frequently had moderate to severe renal disease (11% vs. 7%; $P < 0.001$). Inferior MI was the dominant localization among patients presenting HAVB, accounting for 71% of cases. Eleven percent of patients with HAVB on admission had pre-hospital cardiac arrest. Multi-vessel disease was more frequent among HAVB than in patients without HAVB (66% vs. 61%; $P = 0.019$). PCI was performed more frequently in HAVB patients (87% vs. 84%; $P = 0.012$).

In a multiple logistic regression model higher age, female sex, STEMI, Killip class > 2, inferior as well as posterior ischaemic region, diabetes mellitus and current smoking were identified as independent predictors for HAVB at hospital admission (*Table 2*).

Outcome

Patients presenting with HAVB on admission had higher in-hospital mortality than those without HAVB (15% vs. 4.9%; $P < 0.001$). After multivariable adjustment, HAVB was identified as independent predictor of in-hospital death (OR 1.89, 95% CI, 1.42–2.49; $P < 0.001$) (*Table 3*). The effect of HAVB on in-hospital mortality was comparable between men and women (OR 1.03; 95% CI, 0.68–1.55; $P_{interaction} = 0.895$). Regarding AMI localization, the adverse effect of HAVB on mortality was more pronounced in anterior AMI, compared to patients with non-anterior MI (OR 3.69; 95% CI, 1.99–6.81; $P_{interaction} = 0.02$) (*Figure 2*).

The rate of in-hospital MACCE was also higher in HAVB patients compared to patients without HAVB at admission (17% vs. 6.0%; $P < 0.001$). The rate of permanent pacemaker implantation in HAVB on admission patients was 8.0%.

Trend over time

Rates of HAVB and in-hospital mortality were determined for 1-year-intervals (*Figure 3*). The frequency of HAVB remained stable over the study period. Accordingly, no significant difference between the calculated mean frequency of two time periods (2005–2009 and 2020–2024) was noted (1.3% vs. 1.5%; $P = 0.386$). Among patients with HAVB, the in-hospital mortality did not decline significantly over time (22% in 2005% and 16% in 2023: $p_{trend} = 0.756$), while it was dramatically reduced among patients without HAVB, with mortality rates of 6% and 2%, respectively ($p_{trend} < 0.001$). Regardless of the HAVB status, the mortality curve after 2020 shows a trend of steeper decline.

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Table 1 Baseline characteristics of acute myocardial infarction patients according to the presence of high-grade atrioventricular block at admission

Characteristic	n	Overall, n = 50 279[a]	HAVB at admission: No, n = 49 532[a]	HAVB at admission: Yes, n = 747[a]	P-value
Age, years	50 279	66.5 (56.6, 76.6)	66.4 (56.5, 76.6)	72.0 (61.7, 80.1)	**<0.001**[b]
Female	50 279	12 984 (25.8)	12 716 (25.7)	268 (35.9)	**<0.001**[c]
Diagnosis	50 279				**<0.001**[c]
STEMI		28 780 (57.2)	28 208 (56.9)	572 (76.6)	
NSTEMI		21 499 (42.8)	21 324 (43.1)	175 (23.4)	
Risk factors					
Current smoker	43 820	16 744 (38.2)	16 489 (38.2)	255 (40.9)	0.169[c]
Hypertension	47 440	30 397 (64.1)	29 952 (64.1)	445 (65.1)	0.589[c]
Dyslipidemia	44 838	28 608 (63.8)	28 220 (63.9)	388 (59.8)	**0.032**[c]
Diabetes mellitus	47 366	10 050 (21.2)	9869 (21.1)	181 (26.3)	**0.001**[c]
Obesity (BMI > 30 kg/m^2)	45 054	9619 (21.3)	9501 (21.4)	118 (17.6)	**0.018**[c]
Comorbidities					
History of myocardial infarction	49 284	8181 (16.6)	8076 (16.6)	105 (14.6)	0.156[c]
History of ischaemic heart diseases	41 040	11 688 (28.5)	11 544 (28.6)	144 (23.3)	**0.004**[c]
Cardiac insufficiency	48 743	1467 (3.0)	1449 (3.0)	18 (2.5)	0.434[c]
Peripheral vascular disease	49 242	2573 (5.2)	2525 (5.2)	48 (6.7)	0.075[c]
Cerebrovascular disease	49 242	2713 (5.5)	2669 (5.5)	44 (6.1)	0.458[c]
Moderate to severe renal disease	48 885	3657 (7.5)	3580 (7.4)	77 (10.8)	**<0.001**[c]
CCI > 1	48 385	11 071 (22.9)	10 878 (22.8)	193 (27.1)	**0.007**[c]
Regular medication					
Aspirin	45 880	16 049 (35.0)	15 829 (35.0)	220 (33.4)	0.387[c]
Beta blocker	45 499	13 027 (28.6)	12 866 (28.7)	161 (25.0)	**0.038**[c]
ACE inhibitors/angiotensin II antagonist	45 675	17 380 (38.1)	17 122 (38.0)	258 (39.8)	0.368[c]
Calcium channel blocker	45 073	6735 (14.9)	6622 (14.9)	113 (17.6)	0.057[c]
Maximal creatine kinase, IU/L	46 632	685.0 (240.0, 1787.0)	680.0 (238.0, 1777.0)	1144.0 (412.0, 2314.0)	**<0.001**[b]
Vital parameters					
Systolic blood pressure, mm Hg	48 194	135.0 (118.0, 155.0)	136.0 (119.0, 155.0)	111.0 (91.0, 136.0)	**<0.001**[b]
Diastolic blood pressure, mm Hg	47 777	80.0 (70.0, 90.0)	80.0 (70.0, 90.0)	63.0 (50.0, 77.0)	**<0.001**[b]
Heart rate, beats per minute	48 213	76.0 (65.0, 89.0)	77.0 (66.0, 89.0)	54.0 (43.3, 70.8)	**<0.001**[b]
Left ventricular ejection fraction_impaired	42 439	3754 (8.8)	3691 (8.8)	63 (10.0)	0.304[c]
Killip class	49 616				**<0.001**[c]
Class I		41 658 (84.0)	41 160 (84.2)	498 (67.0)	
Class II		4355 (8.8)	4282 (8.8)	73 (9.8)	
Class III		1372 (2.8)	1349 (2.8)	23 (3.1)	
Class IV		2231 (4.5)	2082 (4.3)	149 (20.1)	
Symptoms					
Pain	48 662	44 112 (90.6)	43 546 (90.8)	566 (80.4)	**<0.001**[c]
Dyspnea	44 423	14 118 (31.8)	13 877 (31.7)	241 (37.2)	**0.003**[c]
Localization ischaemic region					
Anterior	49 363	17 371 (35.2)	17 274 (35.5)	97 (13.2)	**<0.001**[c]
Inferior	49 334	17 288 (35.0)	16 764 (34.5)	524 (70.8)	**<0.001**[c]
Posterior	48 783	4394 (9.0)	4267 (8.9)	127 (17.3)	**<0.001**[c]
Lateral	36 187	5673 (15.7)	5598 (15.7)	75 (13.3)	0.113[c]
Undetermined	49 254	10 666 (21.7)	10 571 (21.8)	95 (13.0)	**<0.001**[c]

Continued

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Table 1 *Continued*

Characteristic	n	Overall, n = 50 279[a]	HAVB at admission		P-value
			No, n = 49 532[a]	Yes, n = 747[a]	
Immediate treatment					
Pre-hospital resuscitation	50 278	2851 (5.7)	2766 (5.6)	85 (11.4)	**<0.001**[c]
Pharmacological					
Aspirin	49 248	47 211 (95.9)	46 501 (95.9)	710 (95.9)	0.910[c]
P2Y12-receptor inhibitor	49 198	42 140 (85.7)	41 512 (85.7)	628 (85.0)	0.598[c]
β-Blocker	48 944	28 037 (57.3)	27 842 (57.8)	195 (26.6)	**<0.001**[c]
Interventional					
PCI	49 300	41 260 (83.7)	40 614 (83.6)	646 (87.1)	**0.012**[c]
Door to balloon time for PCI, min	36 407	75.0 (27.0, 263.0)	75.0 (28.0, 268.0)	47.0 (24.0, 95.8)	**<0.001**[b]
Culprit vessel	25 626				**<0.001**[c]
Left main		625 (2.4)	617 (2.4)	8 (1.9)	
Left anterior descending artery (or one of its branches)		10 912 (42.6)	10 875 (43.1)	37 (8.9)	
Left circumflex artery (or one of its branches)		4626 (18.1)	4597 (18.2)	29 (7.0)	
Right coronary artery (or one of its branches)		8817 (34.4)	8481 (33.6)	336 (80.8)	
Other		646 (2.5)	640 (2.5)	6 (1.4)	
Multi-vessel disease	44 431	27 291 (61.4)	26 840 (61.4)	451 (65.7)	**0.019**[c]
Pacing					
Temporary pacing	48 862	1153 (2.4)	891 (1.9)	262 (35.5)	**<0.001**[c]
Permanent pacemaker implantation	48 742	294 (0.6)	235 (0.5)	59 (8.0)	**<0.001**[d]
Complication					
Re-infarction	49 336	416 (0.8)	407 (0.8)	9 (1.2)	0.267[c]
Cardiogenic shock	49 332	1618 (3.3)	1540 (3.2)	78 (10.5)	**<0.001**[c]
Cerebrovascular event	49 336	374 (0.8)	360 (0.7)	14 (1.9)	**<0.001**[c]
Acute renal failure (needing dialysis)	49 216	817 (1.7)	780 (1.6)	37 (5.0)	**<0.001**[c]
Sepsis/SIRS/multiorgan failure	49 216	643 (1.3)	616 (1.3)	27 (3.6)	**<0.001**[c]
Atrial fibrillation at discharge	49 216	567 (1.2)	546 (1.1)	21 (2.8)	**<0.001**[c]
New heart failure (Killip class III-IV)	49 216	642 (1.3)	619 (1.3)	23 (3.1)	**<0.001**[c]
Outcome					
Hospital duration, days	49 915	4.0 (2.0, 7.0)	4.0 (2.0, 7.0)	5.0 (2.0, 9.0)	**<0.001**[b]
MACCE	49 336	3043 (6.2)	2917 (6.0)	126 (17.0)	**<0.001**[c]
Death	50 279	2530 (5.0)	2415 (4.9)	115 (15.4)	**<0.001**[c]

Significant values in bold.
[a]Median (IQR) *n* (%).
[b]Wilcoxon rank sum test.
[c]Pearson's Chi-squared test.
[d]Fisher's exact test.
ACE, angiotensin-converting enzyme; BMI, body mass index; CCI, Charlson comorbidity index; IQR, interquartile range; MACCE, major adverse cardiac and cerebrovascular event; NSTEMI, non-ST-elevation myocardial infarction; PCI, percutaneous coronary intervention; SIRS, systemic inflammatory response syndrome; STEMI, ST-elevation myocardial infarction.

STEMI vs. NSTEMI in patients presenting with HAVB

The frequency of HAVB on admission in patients with STEMI and NSTEMI were 2% and 0.8% ($P < 0.001$), respectively. Patients presenting with HAVB at hospital admission were compared with respect to the final diagnosis, NSTEMI vs. STEMI. NSTEMI patients with HAVB were ~4 years older and more frequent males (72% vs. 62%; $P = 0.013$), presented less often with typical symptoms such as chest-pain (67% vs. 84%; $P < 0.001$) and with a rate of diabetes almost two times higher (41% vs. 22%; $P < 0.001$) than HABV patients presenting with STEMI. HABV patients with NSTEMI showed higher rates of previous ischaemic heart diseases (33% vs. 20%; $P = 0.001$), a higher CCI score (CCI > 1: 34% bs. 25%; $P = 0.026$) as well as higher rate of multivessel disease (75% vs. 63%; $P = 0.010$).

In terms of in-hospital mortality (15% vs. 16%; $P = 0.800$) and MACCE (16% vs. 19%; $P = 0.402$), both groups were comparable. In

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Table 2 Predictors of high-grade atrioventricular block at admission in patients with acute myocardial infarction

	HAVB		
Predictors	***Odds ratios***	***CI***	***P***
Age, per year	1.04	1.03–1.05	**<0.001**
Female sex	1.20	1.00–1.44	**0.049**
NSTEMI	0.51	0.41–0.63	**<0.001**
Killip > 2	3.74	2.92–4.75	**<0.001**
Resuscitation prior admission	1.23	0.88–1.69	0.221
Ischaemic region: anterior	0.45	0.34–0.60	**<0.001**
Ischaemic region: inferior	2.95	2.38–3.68	**<0.001**
Ischaemic region: posterior	1.40	1.12–1.75	**0.003**
Diabetes mellitus	1.25	1.02–1.53	**0.033**
CCI > 1	0.88	0.71–1.09	0.247
Smoking	1.41	1.17–1.71	**<0.001**
Observations		40352	

Significant values in bold.

Table 3 Independent predictors for death in acute myocardial infarction

	Death		
Predictors	***Odds ratios***	***CI***	***P***
HAVB	1.89	1.42–2.49	**<0.001**
Age, per year	1.06	1.05–1.06	**<0.001**
Female sex	1.09	0.97–1.22	0.133
NSTEMI	0.56	0.50–0.63	**<0.001**
Killip class > 2	5.77	5.13–6.49	**<0.001**
Resuscitation prior admission	6.73	5.81–7.79	**<0.001**
Ischaemic region: anterior	1.17	1.03–1.32	**0.014**
Ischaemic region: inferior	0.97	0.86–1.10	0.673
Ischaemic region: posterior	1.21	1.01–1.45	**0.041**
Diabetes mellitus	1.21	1.08–1.36	**0.001**
CCI > 1	1.85	1.65–2.08	**<0.001**
PCI	0.37	0.33–0.42	**<0.001**
Observations		44224	

Significant values in bold.

the setting of HAVB, NSTEMI patients required more frequently permanent pacemaker implantation compared to the STEMI patients (20% vs. 4%; $P < 0.001$). Data are illustrated in *Figure 4* and Supplementary material online, *Table S2*.

HAVB ad admission vs during hospitalisation

Regarding the timing of onset of HAVB, this was detected in 747 patients (1.5%) at hospital admission and in 407 patients (0.8%) during hospitalization. Patients with HAVB at admission had lower blood pressure and heart rate than those who had HAVB during hospitalization and presented with higher Killip Class (see Supplementary material online, *Table S1*).

Conversely, patients with documented HAVB during hospitalization had more extensive myocardial damage, as shown by a greater proportion of patients with impaired LV-Function (17% vs. 10%; $P < 0.001$), higher median peaks of creatine kinase(1331 vs. 1144; $P = 0.005$), a higher incidence of anterior MI (21% vs. 13%; $P < 0.001$) and cardiogenic shock (20% vs. 11%; $P < 0.001$). Patients who developed HAVB during hospital stay had higher in-hospital mortality (20% vs. 15%; $P = 0.031$), MACCE (25% vs. 17%; $P = 0.001$), as well as higher rate of permanent pacemaker implantation (25% vs. 8%; $P < 0.001$) than patients with HAVB on admission.

Discussion

To the best of our knowledge, this study describes the largest clinical cohort of patients with HAVB related to AMI in the PCI-era. The main findings of our study are the following:

- Although HAVB is a rare complication of AMI, it remains associated with high mortality and MACCE, particularly in the setting of anterior AMI or when it occurs during hospitalization. In addition, HAVB was found to be an independent predictor of in-hospital mortality and adverse outcomes. While a major reduction in mortality was observed over nearly 20 years in patients not having HAVB, this was not the case for patients presenting with HAVB until 2020. Interestingly, thereafter a steeper decline in mortality was noted. The reasons remain hypothetical, but this trend could be explained by the broader adoption of percutaneous support devices, such as Impella, which have recently been shown to reduce mortality in unstable patients with AMI.[24]
- While HAVB was more frequent in patients with STEMI than with NSTEMI, the rate of permanent pacemaker implantation was higher among NSTEMI patients with this condition. The occurrence of HAVB during hospitalization was associated with higher mortality compared with HAVB on admission.

Frequency and predictors of HAVB at admission in AMI

The rate of HAVB at admission in our cohort (1.5%) was slightly lower compared to previous analyses reporting incidences in the range of 2.6%[3] to 2.7%.[4,23] In the study by Kim *et al.*,[23] patients were sicker, with a higher cardiovascular disease burden and a higher Killip class compared to our cohort whereas the studies by Alnsasra *et al.*,[3,4] consisting in a cohort with a higher proportion of diabetes and lower rate of PCI in the early phase of the study considered HAVB on admission and occurring during hospital stay as a cumulative event during the entire hospitalization. Overall, HAVB in the clinical setting of AMI remains a rare event.

Of note, we identified, among others, female sex as independent risk factor for HAVB at hospital admission. This is in accordance with previous observations,[2,3,7,20] where a clear explanation could not be detected.

Impact of HAVB at admission on in-hospital mortality and MACCE

Our data, identifying HAVB as an independent predictor of in-hospital death, mirror previous results by Santos *et al.*[2] in a smaller cohort of HAVB patients in the setting of AMI ($n = 626$). The impact of HAVB on in-hospital mortality was though greater in the cohort studied by Santos *et al.* compared to our findings, namely a mortality rate of

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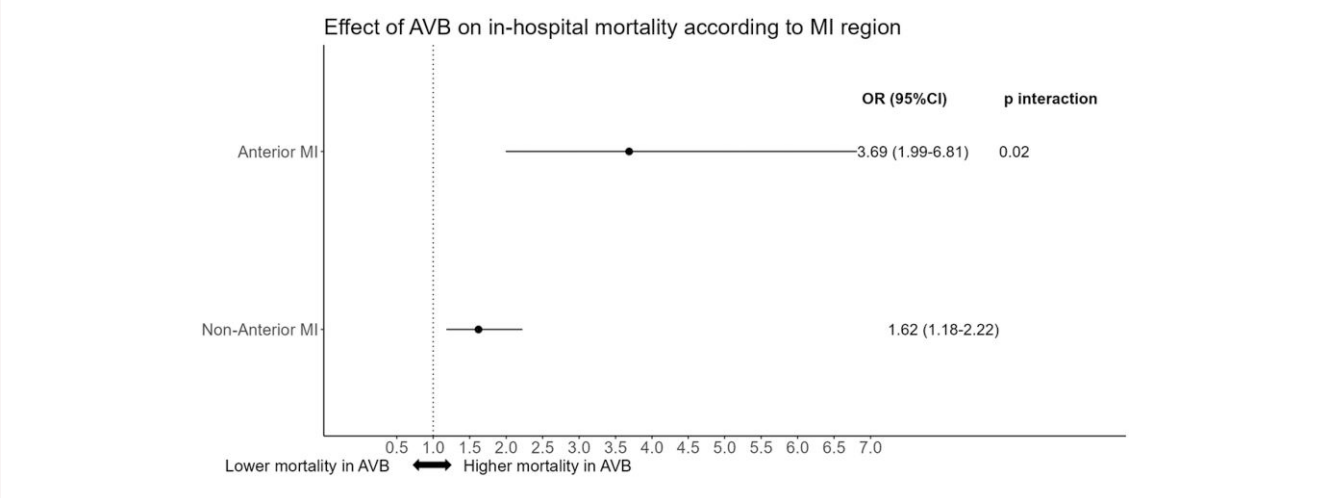

Figure 2 Impact of high-grade atrioventricular blocks on mortality regarding the localization of infarction.

19% vs.15% (OR 2.69 vs. OR 1.89). Of note, Santos *et al.* included any HAVB (on admission or during hospital stay) to this analysis which might explain the higher mortality observed. In our study we differentiated between HAVB on admission and HAVB during hospitalization. Also, the unfavourable association between HAVB complicating anterior AMI and high in-hospital mortality was previously reported only in two other analyses.[2,20]

In our study, mortality rate and MACCE was comparable between HAVB STEMI and NSTEMI patients; however, NSTEMI patients were significantly older, presented with more cardiovascular risk factors and above all with a more diffuse coronary artery disease, all factors predicting *per se* a higher mortality, regardless of the HAVB status. Similar conclusions were reported, however, without a direct comparison in the study by Alnsasra *et al.*[4] In contrast, analysing the correlation between the clinical presentation and in-hospital mortality, Singh *et al.*,[6] in a study of 1701 AMI patients with HAVB, reported an OR of 8.2 for unstable angina, 6.4 for NSTEMI and 3.0 for STEMI with $p_{interaction} < 0.001$. Taking together, this suggests that HAVB is the result of a more complex multi-vessel ischaemia with limited efficacy of reperfusion in NSTEMI compared with the more often single vessel occlusion responding well to reperfusion in STEMI.

HAVB on admission vs. HAVB occurring during hospitalisation

Our study showed that patients with HAVB occurring during hospitalization presented with larger MI (as suggested by a higher rate of patients with impaired LV-function and higher CK-peaks), a higher proportion of anterior MI, a higher rate of cardiogenic shock resulting in a higher mortality rate, longer hospital stay and higher rate of permanent pacemaker implantation. This aspect has been addressed in only one previous study. Comparing acute coronary syndrome patients with HAVB at admission to patients where the heart block was only registered during hospitalization, Singh *et al.*[6] reported similar results to our own, namely larger MI, a higher mortality rate (29% vs. 15%; OR = 2.9; 95% CI = 2.1–4.0) and a higher rate of permanent pacemaker implantation (7.3% vs. 4.2%; $P = 0.01$) in patients with HAVB during hospital stay. In conclusion, the occurrence of HAVB during hospital stay is associated with higher in-hospital mortality, likely as the expression of a more extensive myocardial injury.

Rate of permanent pacemaker implantation in NSTEMI vs STEMI

An original observation from our study, not previously reported, was the nearly five-fold higher rate of permanent pacemaker implantation among HAVB-NSTEMI patients compared to the HAVB STEMI group.

The reasons for the observed difference can only be postulated. In our analysis, we observed that NSTEMI patients present more markers of systemic compromise (older age, a higher rate of diabetes, and elevated comorbidity indices) as well as indicators of more advanced coronary artery disease (a markedly higher history of coronary disease). They also experienced longer delays to revascularization and had lower overall revascularization rates. Taken together, these factors point to greater fragility of the cardiac conduction system, potentially due to pre-existing myocardial fibrosis, as previously reported in the literature. Histopathological findings from Song *et al.*[25] showed significant tissue remodelling, with increased fibrous and fatty tissue in the cardiac conduction system, which increases with aging, while a recent report by Zhang *et al.*[26] demonstrated that diabetes mellitus and advanced age, among other factors, are independently associated with high-degree AVB. Furthermore, we know that patients with extreme bradycardia, such as those with HAVB, can develop elevated troponin levels without necessarily presenting a coronary obstruction.[27–29] The high rate of undetermined culprit lesions in the NSTEMI group (40% of cases) suggests that a portion of these patients may have experienced a primary rhythm disturbance and, consequently, a type II myocardial infarction. Unfortunately, based on our data, we cannot draw definitive conclusions. Further investigation is warranted to clarify this critical issue, which may carry important therapeutic implications.

Strengths and limitations of this study

The main strengths include the sample size since, to the best of our knowledge, this study describes the largest clinical cohort of patients with HAVB related to AMI. In addition, this is one of the first studies to compare in-hospital outcome in subgroups of HAVB patients (i.e. STEMI vs. NSTEMI and anterior vs. non-anterior AMI). The long period of inclusion (20 years) is a strength because it allows to study trends over time but at the same time bias may result from changes in diagnosis and treatment. As additional limitations, mortality rates may be underestimated as patients not surviving the pre-hospital phase (e.g. due to bradycardic cardiac arrest) were not included in the registry. In the

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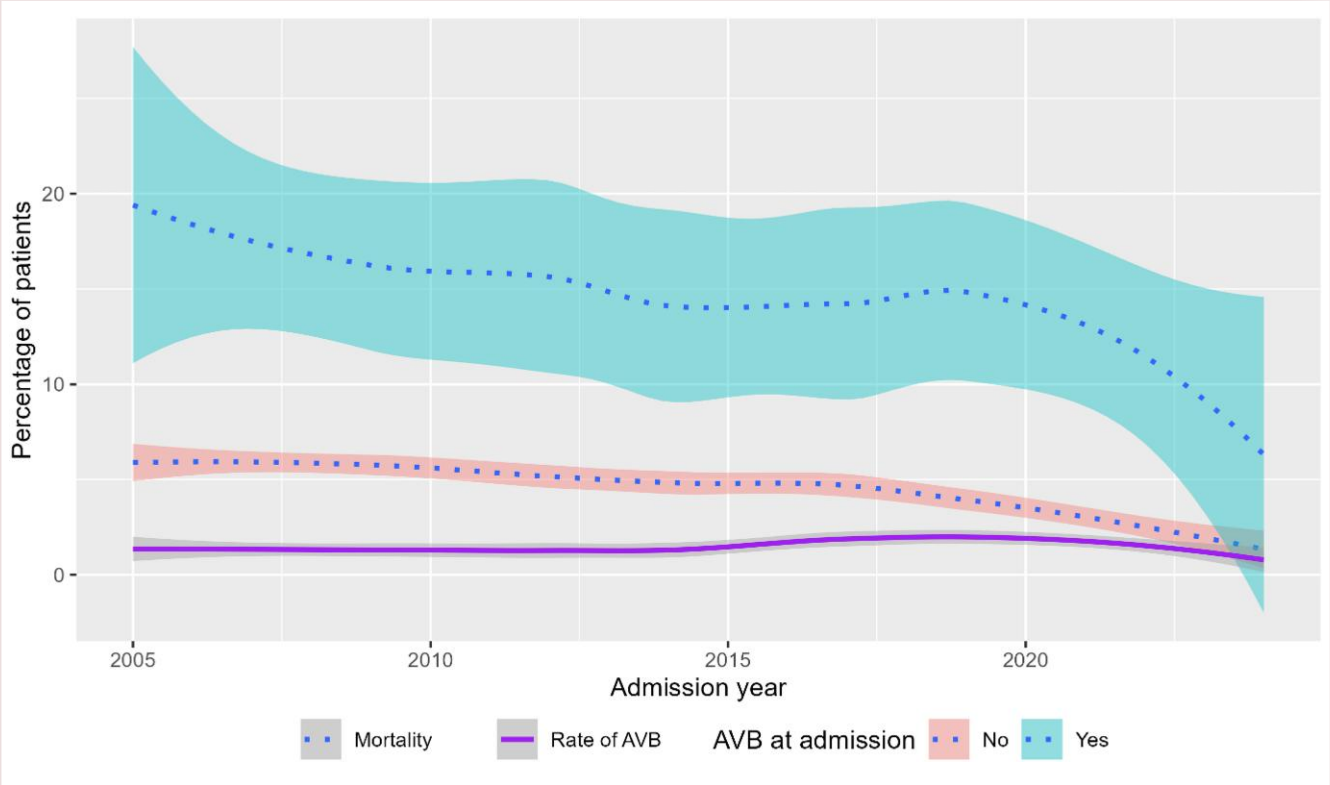

Figure 3 Mortality rate among acute myocardial infarction patients with and without high-grade atrioventricular blocks and frequency of high-grade atrioventricular blocks in acute MI over time.

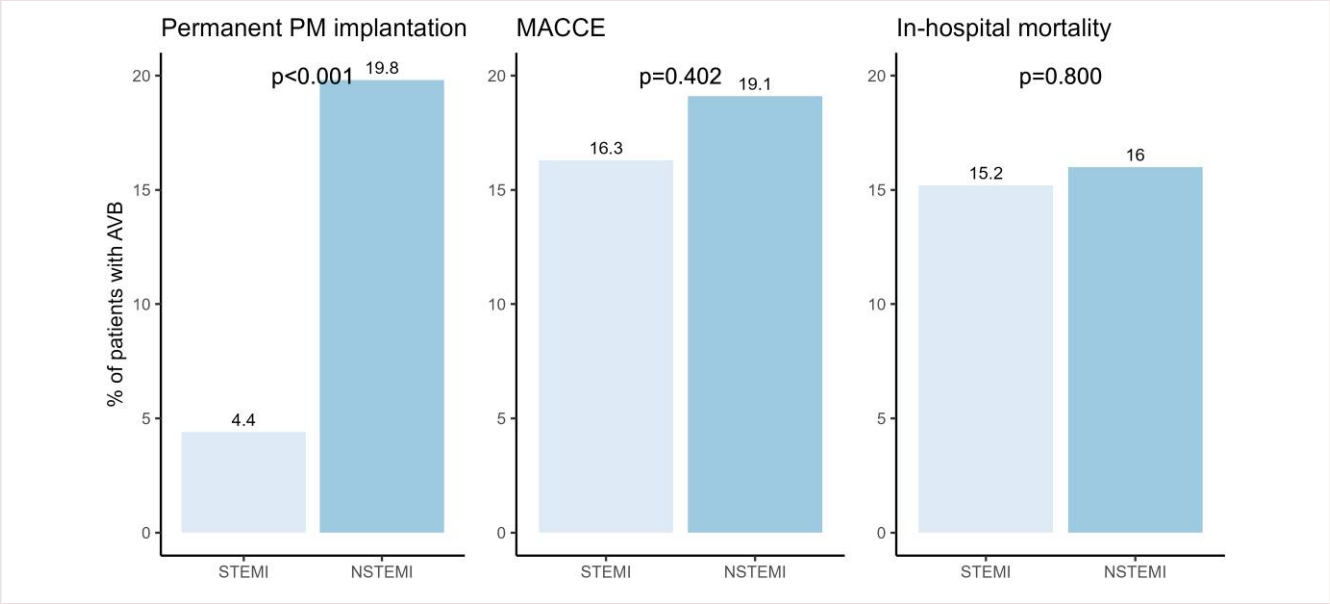

Figure 4 Outcome of acute myocardial infarction patients complicated by high-grade atrioventricular blocks; ST-elevation myocardial infarction vs. non-ST-elevation myocardial infarction.

present study patients with HAVB on admission could either represent persisting or new-onset heart blocks because of lacking systematic access to previous health records. Our dataset further lacked information on HAVB resolution after PCI and its persistence beyond 48 h after revascularization. This prevented conclusive analysis on the association of persistent HAVB and adverse outcomes. Future studies should

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investigate these temporal dynamics and their clinical impact. Complete door-to-balloon time data were available for only a subset of patients, preventing a sufficiently powered multivariable analysis of reperfusion time and in-hospital complications. Finally, data on specific PCI strategies were not available, preventing analysis of their potential impact, particularly in high-risk patients such as those with HAVB.

Conclusion

The frequency of HAVB during the PCI-era remained stable in our cohort, while the in-hospital mortality rate declined in the last years, but HAVB still was an independent predictor for worse in-hospital outcomes. High-grade atrioventricular blocks in acute MI was associated with high in-hospital mortality, especially in anterior AMI or occurring during hospitalization. While the mortality in patients without HAVB significantly decreased over 20 years, this was not the case in patients with HAVB. HAVB in the setting of NSTEMI more frequently required permanent pacemaker implantation. Although data on quality of life over time were not available in our study, future research should address these trends, as they may provide important insights for tailoring long-term patient management and improving clinical practice.

Supplementary material

Supplementary material is available at *European Heart Journal: Acute Cardiovascular Care* online.

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Author contributions

Nicola Olivito (N.O.), Fabienne Foster Witassek (F.F.W.), Hans Rickli (H.R., Prof. Dr. med.), Marco Roffi (M.R., Prof. Dr. med.), Dragana Radovanovic (D.R., Dr.med.), and Giovanni Pedrazzini (G.P., Prof. Dr. med.): The authors N.O., D.R., and G.P. contributed to the conception on the work. All authors contributed to the acquisition of the data. F.F.W. and D.R. performed the statistical analysis. All authors contributed to the interpretation ot data. N.O. drafted the manuscript. All authors critically revised the work and gave a final approval and agreed to be accountable for all aspects of the work, ensuring integrity and accuracy.

Conflict of interest: Marco Roffi (Prof. Dr. med.): Institutional research grants from Cordis, Boston Scientific, Terumo, Biotronik and Vascular Medical.

Data availability

The authors confirm that, for approved reasons, some access restrictions apply to the data underlying the findings. Individual data used for the construction of the AMIS Plus registry are property of the hospitals participating in the AMIS registry and may only be made available by each hospital's PI and steering committee. This also applies to derivatives such as the analysis files used for this study. However, after approval of the AMIS Plus steering committee and subsequent negotiation of an individual AMIS plus module contract with the AMIS Plus steering committee, analysis files can be handed over to other researchers. Requests must be submitted to Prof. Dr. Hans Rickli, President of the AMIS Plus Steering Committee (hans.rickli@kssg.ch) and Dr. Dragana Radovanovic (Head of Data Center AMIS Plus, dragana.radovanovic@uzh.ch).

References

1. Simons GR, Sgarbossa E, Wagner G, Califf RM, Topol EJ, Natale A. Atrioventricular and intraventricular conduction disorders in acute myocardial infarction: a reappraisal in the thrombolytic era. *PACE Pacing Clin Electrophysiol* 1998;**21**:2651–2663.
2. Santos H, Santos M, Almeida I, Paula SB, Chin J, Almeida S, *et al.* High-grade atrioventricular block in acute coronary syndrome: portuguese experience. *J Electrocardiol* 2021;**68**:130–134.
3. Alnsasra H, Tsaban G, Weinstein JM, Nasasra M, Ovdat T, Beigel R, *et al.* Sex differences in ventricular arrhythmia, atrial fibrillation and atrioventricular block complicating acute myocardial infarction. *Front Cardiovasc Med* 2023;**10**:1217525.
4. Alnsasra H, Ben-Avraham B, Gottlieb S, Ben-avraham M, Kronowski R, Iakobishvili Z, *et al.* High-grade atrioventricular block in patients with acute myocardial infarction. Insights from a contemporary multi-center survey. *J Electrocardiol* 2018;**51**:386–391.

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5. Misumida N, Ogunbayo GO, Kim SM, Abdel-Latif A, Ziada KM, Elayi CS. Frequency and significance of high-degree atrioventricular block and sinoatrial node dysfunction in patients with non-ST-elevation myocardial infarction. *Am J Cardiol* 2018;**122**:1598–1603.
6. Singh SM, FitzGerald G, Yan AT, Brieger D, Fox KAA, López-Sendón J, *et al.* High-grade atrioventricular block in acute coronary syndromes: insights from the global registry of acute coronary events. *Eur Heart J* 2015;**36**:976–983.
7. Gang UJO, Hvelplund A, Pedersen S, Iversen A, Jons C, Abildstrom SZ, *et al.* High-degree atrioventricular block complicating ST-segment elevation myocardial infarction in the era of primary percutaneous coronary intervention. *Europace* 2012;**14**:1639–1645.
8. AMIS Plus - National Registry of Acute Myocardial Infarction [Internet]. [cited 2024 Nov 11]. Available from: https://amis-plus.ch/
9. Thygesen K, Alpert JS, White HD. Universal definition of myocardial infarction. *Eur Heart J* 2007;**28**:2525–2538.
10. Alpert JS, Antman E, Apple F, Armstrong PW, Bassand JP, De Luna AB, *et al.* Myocardial infarction redefined–a consensus document of the joint European society of cardiology/American College of Cardiology committee for the redefinition of myocardial infarction. *Eur Heart J* 2000;**21**:1502–1513.
11. Thygesen K, Alpert JS, Jaffe AS, Chaitman BR, Bax JJ, Morrow DA, *et al.* Fourth universal definition of myocardial infarction (2018). *Eur Heart J* 2019;**40**:237–269.
12. Thygesen K, Alpert JS, Jaffe AS, Simoons ML, Chaitman BR, White HD, *et al.* Third universal definition of myocardial infarction. *Eur Heart J* 2012;**33**:2551–2567.
13. Killip T, Kimball JT. Treatment of myocardial infarction in a coronary care unit. A two year experience with 250 patients. *Am J Cardiol* 1967;**20**:457–464.
14. Radovanovic D, Seifert B, Urban P, Eberli FR, Rickli H, Bertel O, *et al.* Validity of charlson comorbidity Index in patients hospitalised with acute coronary syndrome. Insights from the nationwide AMIS plus registry 2002–2012. *Heart* 2014;**100**:288–294.
15. Charlson ME, Pompei P, Ales KL, MacKenzie CR. A new method of classifying prognostic comorbidity in longitudinal studies: development and validation. *J Chronic Dis* 1987;**40**:373–383.
16. Hunziker L, Radovanovic D, Jeger R, Pedrazzini G, Cuculi F, Urban P, *et al.* Twenty-year trends in the incidence and outcome of cardiogenic shock in AMIS plus registry. *Circ Cardiovasc Interv* 2019;**12**:e007293.
17. Radovanovic D, Erne P, Pedrazzini G, Stauffer JC, Jeger R, Essig M, *et al.* Temporal trends for cerebrovascular event in patients admitted with acute coronary syndrome 1997 -2012. *Eur Heart J* 2013;**34**:P382–P382.
18. Radovanovic D, Maurer L, Bertel O, Witassek F, Urban P, Stauffer JC, *et al.* Treatment and outcomes of patients with recurrent myocardial infarction: a prospective observational cohort study. *J Cardiol* 2016;**68**:498–503.
19. Kosmidou I, Redfors B, Dordi R, Dizon JM, McAndrew T, Mehran R, *et al.* Incidence, predictors, and outcomes of high-grade atrioventricular block in patients with ST-segment elevation myocardial infarction undergoing primary percutaneous coronary intervention (from the HORIZONS-AMI trial). *Am J Cardiol* 2017;**119**:1295–1301.
20. Kim KH, Jeong MH, Ahn Y, Kim YJ, Cho MC, Kim W, *et al.* Differential clinical implications of high-degree atrioventricular block complicating ST-segment elevation myocardial infarction according to the location of infarction in the era of primary percutaneous coronary intervention. *Korean Circ J* 2016;**46**:315–323.
21. Auffret V, Loirat A, Leurent G, Martins RP, Filippi E, Coudert I, *et al.* High-degree atrioventricular block complicating ST segment elevation myocardial infarction in the contemporary era. *Heart* 2016;**102**:40–49.
22. Gómez-Talavera S, Vivas D, Perez-Vizcayno MJ, Hernández-Antolín R, Fernández-Ortíz A, Bañuelos C, *et al.* Prognostic implications of atrio-ventricular block in patients undergoing primary coronary angioplasty in the stent era. *Acute Card Care* 2014;**16**:1–8.
23. Kim HL, Kim SH, Seo JB, Chung WY, Zo JH, Kim MA, *et al.* Influence of second- and third-degree heart block on 30-day outcome following acute myocardial infarction in the drug-eluting stent era. *Am J Cardiol* 2014;**114**:1658–1662.
24. Møller JE, Engstrøm T, Jensen LO, Eiskjær H, Mangner N, Polzin A, *et al.* Microaxial flow pump or standard care in infarct-related cardiogenic shock. *N Engl J Med* 2024;**390**:1382–1393.
25. Song Y, Laaksonen H, Saukko P, Toivonen S, Zhu J. Histopathological findings of cardiac conduction system of 150 Finns. *Forensic Sci Int* 2001;**119**:310–317.
26. Zhang J, Liu J, Ye M, Zhang M, Yao F, Cheng Y. Incidence and risk factors associated with atrioventricular block in the general population: the atherosclerosis risk in communities study and cardiovascular health study. *BMC Cardiovasc Disord* 2024;**24**:509.
27. Neumann JT, Twerenbold R, Ojeda F, Sörensen NA, Chapman AR, Shah ASV, *et al.* Application of high-sensitivity troponin in suspected myocardial infarction. *N Engl J Med* 2019;**380**:2529–2540.
28. Shah ASV, Anand A, Strachan FE, Ferry AV, Lee KK, Chapman AR, *et al.* High-sensitivity troponin in the evaluation of patients with suspected acute coronary syndrome: a stepped-wedge, cluster-randomised controlled trial. *Lancet* 2018;**392**:919–928.
29. Reichlin T, Hochholzer W, Bassetti S, Steuer S, Stelzig C, Hartwiger S, *et al.* Early diagnosis of myocardial infarction with sensitive cardiac troponin assays. *N Engl J Med* 2009;**361**:858–867.

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